ISYS3416/ISYS3432 Software Engineering Fundamentals for IT

Tute/Lab Week 1 Objectives

- Introduction to Software Engineering
- Software Development Lifecycle
- Common Processes

I.

a. Explain why the project attributes time, cost and scope are interdependent.
b. Suppose your team has been asked to develop a project within three months and you are told that the project scope cannot be reduced. What options are open to you as the project manager of the team?
c. Consider the experience of Jack.

During Jack's six-month internship he was asked to work on a project to deliver a new piece of software. After a couple of weeks, the project sponsor decided to add some new requirements and Jack happily included them. Two weeks later, the sponsor made some more changes and asked for some new functionality. Again, Jack said that it was no problem. The changes were made. Towards the end of the three months, the sponsor complained that Jack was behind schedule. Jack tried to explain that all the changes meant that there was no way that the software could have been completed to the original timescales. The sponsor was not happy, and Jack was labeled as being "too slow".

What caused the problem above? How could Jack have avoided this?

II.

a. What are the main phases of the software development lifecycle?
b. Assume ABC bank has requested you to develop an online Loan-Approval System. Describe some of the tasks that may be carried out in each of the phases.

III. How do agile processes differ from the traditional waterfall model? Explain which of these processes will be better suited for the following applications justifying your reasons.

a. E-commerce applications where business requirements are constantly changing
b. Software to control a nuclear power plant